



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE (Advanced)

PRE - SCORE

22-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-1 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	C,D	A,B	B,C	A,B,C	A,B,D	A,D	B,D	A,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.50	1.00	0.15	2.81	20.00 to 20.20	9.82 to 9.83		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	C	B	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	B,C,D	A,B,D	A,B,C,D	B,C	B,C,D	A,B,C,D	A,B,C,D	C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	6.00	4.00	5.00	5.00	4.00	4.00		

PART-3 : MATHEMATICS

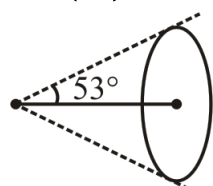
SECTION-I (i)	Q.	1	2	3	4				
	A.	A	D	D	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C,D	A,B,C,D	B,C,D	B,C	B,C	D	B	A,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	0.00	4.00	5.00	4.00	147.00	3.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)



$$P = 56.7\pi \times \frac{2\pi(1 - \cos 53^\circ)}{4\pi}$$

$$= 5.67 \times 10^{-8} \times \pi \times 1^2 \times 2 T^4$$

$$T^4 = \frac{10^9}{2} \times \frac{2}{5} \times 2 = 10^8$$

$$\Rightarrow T = 100 K$$

2. Ans (D)

$$f_{\text{mix}} = \frac{(n\alpha)(3)(3) + (n - n\alpha)(6)}{(n\alpha)(3) + (n - n\alpha)}$$

$$= \frac{3\alpha + 6}{2\alpha + 1}$$

4. Ans (D)

$$\left(P_L + \frac{80^2}{R}\right) t_1 = mL$$

$$\left(P_L + \frac{120^2}{R}\right) t_2 = mL$$

$$\frac{120^2 - 80^2}{R} = mL \left(\frac{1}{4} - \frac{1}{20}\right)$$

$$8000 \times 5 = mLR$$

$$P_L R + 6400 = \frac{8000 \times 5}{20} = 2000$$

$$P_L R = -4400$$

$$\frac{1600 - 4400}{R} t = -ve$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (C,D)

$$V = aT^2$$

$$\Rightarrow PV^{1/2} = \text{constant}$$

$$C = \frac{R}{1-x} + \frac{R}{\gamma-1}$$

$$\frac{R}{1-\frac{1}{2}} + \frac{R}{\frac{5}{3}-1} = \frac{7R}{2}$$

$$V = aT^2$$

$$dV = 2aTdT \quad \dots\dots (1)$$

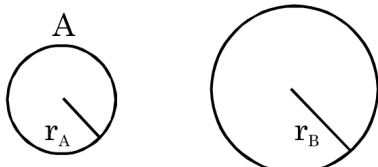
$$P = \frac{nRT}{V} = \frac{nRT}{aT^2} = \frac{nR}{aT}$$

$$PdV = \left(\frac{nR}{aT} \right) 2aTdT = 2nRdT$$

From eqn. (1)

$$PdV = \int_{T_1}^{T_2} 2nRdT = (2nR)(\Delta T) = (2)(1)(R)(100) = 200R$$

6. Ans (A,B)



$$\lambda_A = 400 \text{ nm}$$

$$\lambda_B = 800 \text{ nm}$$

Now $\lambda T = \text{constant}$

$$\Rightarrow \lambda_A T_A = \lambda_B T_B$$

$$\Rightarrow 400T_A = 800T_B$$

$$\Rightarrow T_A = 2T_B$$

$$\Rightarrow \frac{T_A}{T_B} = \frac{2}{1}$$

$\therefore$ (A)

Again P is energy emitted per second

$$\therefore P \propto AT^4$$

$$\Rightarrow \frac{P_A}{P_B} = \frac{A_a}{A_b} \left(\frac{T_a}{T_b} \right)^4 = \frac{r_a^2}{r_b^2} \left(\frac{T_a}{T_b} \right)^4 = \frac{1}{4} \cdot 2^4 = 4$$

$\therefore$ (B)

Again emissive power = $\frac{\text{energy emitted per sec}}{\text{area}}$

$$\Rightarrow E = \frac{P}{A} \Rightarrow E \propto \frac{P}{r^2}$$

$$\Rightarrow \frac{E_A}{E_B} = \left(\frac{P_a}{P_b} \right) \left(\frac{r_b}{r_a} \right)^2 = 4 \cdot 2^2 = 16$$

$\therefore$ (C, D) are wrong

7. Ans (B,C)

$$(T-0) \rho \times V + PV \times 80000 = (50-T) \rho \times 5$$

$$V = \frac{(90-T) \times 5}{5 + 80000} = 200$$

$$(T-0) \times m \times 1 + m \times 80 = 5000 \times 1 \times (50-T)$$

$$m = \frac{5000(50-T)}{T+80}$$

$$= \frac{5000(35)}{95}$$

$$\text{When } T = 15^\circ\text{C}, V_f = \left(5 + \frac{35}{19} \right) \text{ dm}^3$$

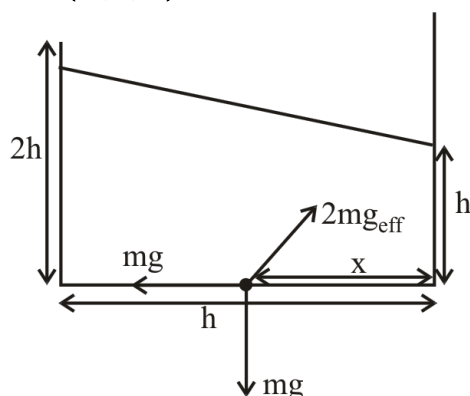
$$\text{When } T = 0, V_f = \left(5 + \frac{25}{8} \right) \text{ dm}^3 + \text{ice may be}$$

present

$$\text{When } T = 5^\circ\text{C}, V_f = \left(5 + \frac{45}{17} \right) \text{ dm}^3$$

$$\text{When } T = 10^\circ\text{C}, V_f = \left(5 + \frac{20}{9} \right) \text{ dm}^3$$

8. Ans (A,B,C)

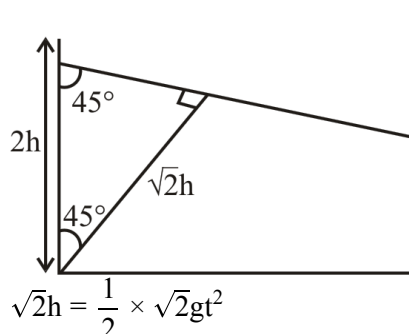


$$\tan \theta = \frac{a}{g} = 1$$

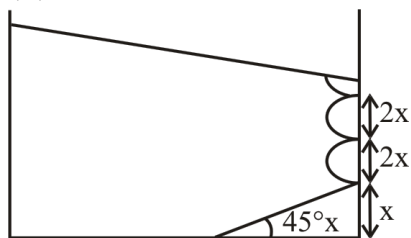
$$a_x = g; a_y = g$$

$$(A) x = 0 \Rightarrow h = \frac{1}{2}gt^2$$

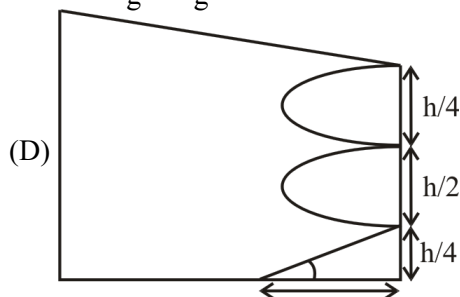
$$(B) x = h$$



(C)



$$S_y = h = \frac{h}{g} + \frac{2h}{g} \times n$$



$$S_y > h \Rightarrow t > \sqrt{\frac{2h}{g}}$$

10. Ans (A,D)

$$T_1 = 2T_2$$

$$A_1 = A_2$$

$$Y_1 = 2Y_2$$

$$\text{Stress} = \frac{T}{A} \text{ \& energy density} = \frac{1}{2} \frac{(\text{Stress})^2}{Y}$$

11. Ans (B,D)

$$(A) T = \sqrt{T_1 T_2}$$

$$= \sqrt{625 \times 289}$$

$$= 25 \times 17 = 425 \text{ K}$$

$$(B) \frac{dT_1}{T_1} = -\frac{dT_2}{T_2}$$

$$\int d\omega = C \int (dT_1 - dT_2)$$

$$\omega = 500 (200 - 136)$$

$$= 3.2 \text{ kJ}$$

$$(C) \omega = 3200 = \int_0^\infty P_0 d\tau$$

$$= 960 \frac{e^{-k\tau}}{-k} \Big|_0^\infty$$

$$k = \frac{960}{3200} = 0.3$$

$$(D) \frac{dw}{dt} = 1 - \frac{289}{625} = \frac{336}{625} = 240$$

12. Ans (A,D)

$$P_0 V_0^{5/3} = P_1 (V_0 - \Delta V) P_0 V_0$$

$$\frac{P_0 V_0}{T_0} = \frac{P_1 (V_0 + \Delta V)}{T_0 + \Delta T}$$

$$\frac{T_0 V_0^{2/3}}{T_0} = \frac{(V_0 - \Delta V)^{5/3}}{V_0 + \Delta V}$$

$$1 + \frac{\Delta T}{T_0} = \left(1 + \frac{\Delta V}{V_0}\right) \times \left(1 - \frac{\Delta V}{V_0}\right)^{-5/3}$$

$$\frac{\Delta T}{T_0} = \frac{8\Delta V}{3V_0}$$

$$T_0 V_0^{2/3} = (T_0 + \Delta T_2) (V_0 - \Delta V)^{2/3}$$

$$1 - \frac{\Delta T_2}{T_0} = \left(1 - \frac{\Delta V}{V_0}\right)^{2/3}$$

$$\frac{\Delta T_2}{T_0} = \frac{2}{3} \frac{\Delta V}{V_0}$$

$$\Rightarrow \Delta T_2 \frac{\Delta T}{4} \Rightarrow Q = \frac{3}{2} R \Delta T + \frac{3}{2} R \Delta T_2 = \frac{15}{8} R \Delta T$$

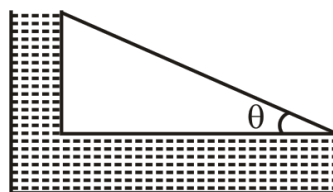
PART-1 : PHYSICS

SECTION-II

1. Ans (2.50)

$$\alpha_1 = \frac{p - p_1}{p_1 (t_1 - t)} + \alpha$$

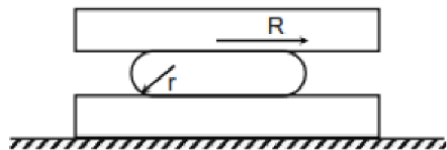
2. Ans (1.00)



$$\tan \theta = \frac{a}{g} = \frac{1}{10}$$

$$a = 1 \text{ m/s}^2$$

3. Ans (0.15)



$$mg = \pi R^2 \Delta P$$

$$\text{Where } \Delta P = S \left(\frac{1}{r} + \frac{1}{R} \right)$$

$$\text{Since } R \gg r$$

$$\therefore \Delta P = S \left(\frac{1}{r} \right)$$

$$\Rightarrow mg = \pi R^2 \frac{S}{r}$$

$$Mg + mg = \pi R'^2 \Delta P'$$

$$\Delta P' = S \left(\frac{1}{r'} + \frac{1}{R'} \right)$$

$$r' = \frac{r}{2}, R' \gg r'$$

$$\therefore \Delta P' = S \left(\frac{2}{r} \right)$$

$$\pi R^2 \cdot 2r = \pi R'^2 \cdot 2r'$$

$$\Rightarrow R' = R\sqrt{2}$$

$$\Rightarrow Mg + mg = \pi (2R^2) S \left(\frac{2}{r} \right)$$

$$4\pi R^2 \frac{S}{r}$$

$$= 4mg \Rightarrow M = 3m$$

$$= 3 \times 0.050 = 0.15 \text{ kg}$$

4. Ans (2.81)

In an inertial frame fixed to the sled, apply

Bernoulli's equation between the scoop (point 1)

and the highest point in the trajectory (point 2)

$$\frac{p_1}{\gamma} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\gamma} + \frac{v_2^2}{2g} + z_2$$

$$h = \frac{v^2 \sin^2 45^\circ}{2g} = \frac{900}{20} \times \frac{1}{16}$$

$$= 2.81 \text{ m}$$

5. Ans (20.00 to 20.20)

$$0.5R = \frac{5R}{2} - \frac{R}{m-1}$$

$$\frac{1}{m-1} = 2$$

$$m = \frac{3}{2}$$

$$4P_0 \times (3V_0)^{3/2} = P \times (4V_0)^{3/2}$$

$$P = P_0 \frac{3\sqrt{3}}{2}$$

$$Q_g = \frac{7}{2} \times (12P_0 V_0 - 4P_0 V_0)$$

$$Q_t = (6\sqrt{3} - 12)P_0 + \frac{5}{2} \times 4V_0 \times V_0 \times \left(P_0 - \frac{3\sqrt{3}}{2} \right) P_0$$

$$\eta = 1 + \frac{(\sqrt{3} - 6 + 10 - 15\sqrt{3} - 8 \times \ln 2)}{2s}$$

6. Ans (9.82 to 9.83)

$$T_S = 0^\circ\text{C}$$

Mas of ice remaining = m

$$mg + 10 \text{ g} = 1 \times \left(\frac{m}{0.9} + \frac{10}{9} \right) \times g$$

$$\frac{m}{g} = 10 - \frac{10}{g} = \frac{80}{9}$$

$$m = 80 \text{ gm}$$

$$\Rightarrow 50 \times 330 \times \frac{10}{10} = 400 \times \frac{4200}{1000} \times (t - 0)$$

$$t = \frac{1650}{168} = 9.82^\circ\text{C}$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (6.00)

i, ii, iii, iv, v, viii

2. Ans (4.00)

a, b, c, d

3. Ans (5.00)

i, ii, iii, v, vi

4. Ans (5.00)

i, ii, iv, vi, vii

5. Ans (4.00)

a, b, c, d

8. Ans (B,C)

$$\frac{\sin \frac{\pi}{12}}{\sin x} + \frac{\cos \frac{\pi}{12}}{\cos x} = 2 \Rightarrow \frac{\sin\left(x + \frac{\pi}{12}\right)}{\sin 2x} = 1$$

9. Ans (B,C)

$$\sin x + \cos x = 1 - a \sin x \cos x$$

$$\Rightarrow a^2 \sin^2 x \cos^2 x - 2(a+1) \sin x \cos x = 0$$

$$\Rightarrow \sin 2x \left(\frac{a^2}{2} \sin 2x - 2(a+1) \right) = 0$$

$$\text{Hence } \sin 2x = 0 \quad \forall a \in \mathbb{R}$$

$$\Rightarrow x = \frac{n\pi}{2}, n \in \mathbb{I}$$

And

$$\sin 2x = \frac{4(a+1)}{a^2} \Rightarrow a \in (-\infty, 2-2\sqrt{2}] \cup [2+2\sqrt{2}, \infty)$$

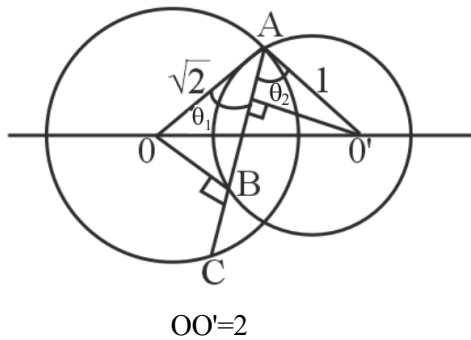
10. Ans (D)

$$\left| \frac{\frac{c}{3}}{\sqrt{\frac{1}{3^2} + \frac{1}{4^2}}} \right| = c \Rightarrow c = 6, 1$$

11. Ans (B)

$$\tan(\theta_1 + \theta_2 + \theta_3) = \frac{\sum \tan \theta_i - \prod \tan \theta_i}{1 - \sum \tan \theta_i \tan \theta_j} = \frac{a+1-b}{1-(b-a)} = 1$$

12. Ans (A,C,D)



$$\text{Let } AC = x \therefore AB = BC = \frac{x}{2} \text{ and } AD = DB = \frac{x}{4}$$

$$\text{in } \triangle OAO' \cos(\theta_1 + \theta_2) = \frac{2+1-4}{2 \cdot \sqrt{2} \cdot 1} = \frac{-1}{2\sqrt{2}} \dots (1)$$

$$\text{also } \sqrt{2} \cos \theta_1 = AB = \frac{x}{2} \Rightarrow \cos \theta_1 = \frac{x}{2\sqrt{2}}$$

$$\text{similarly } 1 \cdot \cos \theta_2 = AD = \frac{x}{4} \Rightarrow \cos \theta_2 = \frac{x}{4}$$

$\therefore$ from equation (1)

$$\frac{x}{2\sqrt{2}} \cdot \frac{x}{4} - \sqrt{1 - \frac{x^2}{8}} \sqrt{1 - \frac{x^2}{16}} = -\frac{1}{2\sqrt{2}}$$

$$\Rightarrow x = \sqrt{\frac{7}{2}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.00)

Let $P(x,0)$

$$\sqrt{(x+2)^2 + 16} + \sqrt{(x+3)^2 + 25} = 7$$

$$\min PA = 4, \min PB = 5$$

Not possible

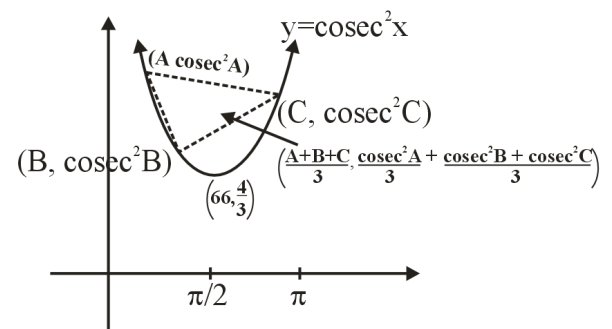
2. Ans (4.00)

$$\therefore \text{in } \triangle ABC \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2} \leq \frac{1}{8}$$

$$\therefore \operatorname{cosec} \frac{A}{2} \cdot \operatorname{cosec} \frac{B}{2} \cdot \operatorname{cosec} \frac{C}{2} \geq 8 \Rightarrow p = 8$$

$$\operatorname{cosec}^2 A + \operatorname{cosec}^2 B + \operatorname{cosec}^2 C > 4 \Rightarrow q = 4$$

$$\therefore p - q = 4$$



$$\operatorname{cosec}^2 A + \operatorname{cosec}^2 B + \operatorname{cosec}^2 C > 4 \Rightarrow q = 4$$

$$\therefore p - q = 4$$

3. **Ans (5.00)**

Since diagonal is a member of both the families so it will pass through (1,0) and (-1,1)

Equation of diagonal AC is $x + 2y - 1 = 0$

Since one of the vertex (3,2) which does not be on

AC, so equation of BD is $2x - y = 4$

Point of intersection of AC and BD is $P \left(\frac{9}{5}, -\frac{2}{5} \right)$

If vertex B is (3,2) then vertex D is $\left(\frac{3}{5}, -\frac{14}{5} \right)$

also $BD = \frac{12\sqrt{5}}{5}$ (say d_1)

Area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$\frac{1}{2} \times \left(\frac{12\sqrt{5}}{5} \right) \times d_2 = 12\sqrt{5}$$

$$d_2 = 10$$

4. **Ans (4.00)**

$$\sin^4 x - \cos^2 x \sin x + 2\sin^2 x + \sin x = 0$$

$$\sin^4 x - (1 - \sin^2 x)\sin x + 2\sin^2 x + \sin x = 0$$

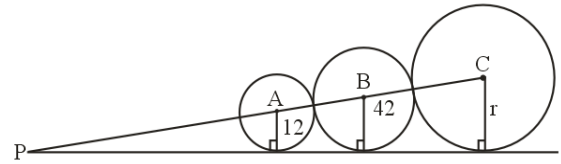
$$\sin^2 x [\sin^2 x + \sin x + 2] = 0$$

$$\sin x = 0 \therefore \sin^2 x + \sin x + 2 > 0 ; x \in \mathbb{R}$$

$$x = 0, \pi, 2\pi, 3\pi$$

5. **Ans (147.00)**

$$\text{We have } \sin \theta = \frac{12}{PA} = \frac{42}{PB} = \frac{r}{PC} = K \text{ (Let)}$$



$$\text{So } K(PB - PA) = 42 - 12 = 30$$

$$K(AB) = 30$$

$$K[54] = 30$$

$$K = \frac{30}{54} = \frac{5}{9}$$

$$\text{Now } K(PC - PB) = r - 42$$

$$K(BC) = r - 42$$

$$K[r + 42] = r - 42$$

$$\text{On solving } r = 147$$

6. **Ans (3.00)**

Denoting the two curve by S_1 and S_2 the equation of curve S passing through intersection of S_1 and S_2 can

be $S = S_1 + \lambda S_2 = 0$. Since S is circle

$\Rightarrow$ Coefficient of $x^2 =$ coefficient of y^2 , coefficient of

$xy = 0$ we get $\lambda = 1$

$$S = (a' + a)(x^2 + y^2) - 2(g' + g)x - 2(f' + f)y + 2c = 0$$

The centre is $\left(\frac{g' + g}{a' + a}, \frac{f' + f}{a' + a} \right)$ which is P

$$PA^2 + PB^2 + PC^2 = 3 (\text{Radius})^2 = 3(PD)^2$$



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(Academic Session : 2023 - 2024)

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PRE - SCORE

22-10-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

ANSWER KEY

PAPER-2 (OPTIONAL)

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,B	B,D	B,C	B,D	A,B,C,D	B,C	C,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	A	B	D				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	330.00	10.00	44.50	2.38	1.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,C	A,D	B,C	A,B	A,B,C	A,C	A,B,C	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	C	C	D	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	5.00	6.00	4.00	6.00	4.00	5.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,B,D	A,B	C,D	A,C,D	A,B,D	A,B,C	B,D	A,B
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	2.00	8.00	2.00	0.00	5.00	25.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,C)

$$P_1 V_1 = P_2 V_2$$

$$10^5 \times 16 = P_2 \times 15$$

$$P_2 = 106.6 \text{ kPa}$$

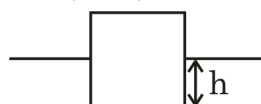
2. Ans (A,B)

$$\frac{dH}{dt} = -kA \frac{dT}{dx}$$

$$(A) A \downarrow \Rightarrow \frac{dT}{dx} \uparrow$$

$$(B) k \downarrow \Rightarrow \frac{dT}{dx} \uparrow$$

3. Ans (B,D)



$$B = mg = \text{const.}$$

$$\frac{h}{h_0} = \frac{1}{2}$$

$$\frac{V_{LD}}{V} = \frac{\rho}{\rho_0} = \frac{1}{2} \frac{(1 + 2\gamma\Delta T)}{(1 + \gamma\Delta T)}$$

$$= \frac{1}{2} (1 + \gamma\Delta T)$$

$$\frac{h_{LD}}{h} = \frac{1}{2} (1 + \gamma\Delta T)$$

$$A = A_0 \left(1 + \frac{2\gamma}{3} \Delta T \right)$$

$$\frac{\Delta A}{A_0} = \frac{2\gamma}{3} \Delta T$$

5. Ans (B,D)

$$V = V_0 + kt$$

$$r \propto (V_0 + kt)^{1/3}$$

$$p = \frac{4S}{(V_0 + kt)^{1/3}}$$

$$\frac{dP}{dt} = -\frac{1}{3} \frac{4S}{(V_0 + kt)^{4/3}}$$

$$E = S_x 4\pi r^2 \propto (V_0 + kt)^{2/3}$$

$$\frac{dE}{dt} = C \times \frac{2}{3} \frac{1}{(V_0 + kt)^{1/3}} \times k$$

6. Ans (A,B,C,D)

$$r > 0.1 \text{ m}, v = 3 - 30(r - 0.1)$$

$$\frac{F}{A} = \eta \frac{dv}{dr}$$

$$\frac{F}{2\pi r \ell} = \eta \frac{dv}{dr}$$

$$\frac{F}{\ell} = 2\pi \times 0.2 \times \frac{3}{0.1} = 12\pi$$

$$\frac{dV}{dt} = \pi \times 0.1^2 \times 3 + 2\pi \int_{0.1}^{0.2} r dr \times (6 - 30r)$$

$$= \frac{3\pi}{100} + 2\pi \left[6 \times \frac{(0.2^2 - 0.1^2)}{2} - \frac{30 \times (0.2^3 - 0.1^3)}{3} \right]$$

$$= \frac{3\pi}{100} + \frac{2\pi}{100} [9 - 7] = \frac{7\pi}{100} \text{ m}^3/\text{s}$$

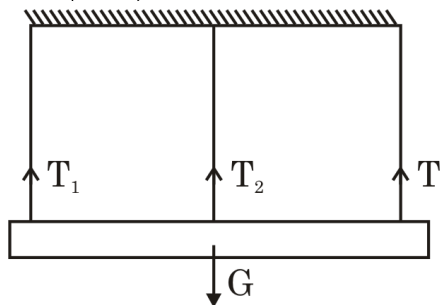
7. Ans (B,C)

$$\frac{dW}{dt} = \rho g \frac{dV}{dt}$$

$$\Rightarrow \frac{dV}{dt} = 8.5 \times 10^{-3} \text{ m}^3/\text{s} = 8500 \text{ cm}^3/\text{s}$$

water is entering $\Rightarrow$ air flows out.

8. Ans (C,D)



$$2T_1 + T_2 = 1000$$

$$T_1 = Y_{cu} A \frac{\Delta \ell}{\ell_0}$$

$$T_2 = Y_s A \frac{\Delta \ell}{\ell_0}$$

$$\Rightarrow T_2 = 2T_1$$

$$T_1 = 250 \text{ N}$$

$$T_2 = 500 \text{ N}$$

PART-1 : PHYSICS

SECTION-I (ii)

9. Ans (C)

$$F_{\text{base}} = 0.1 \times 10^3 \times 10 \times 0.01 = 10 \text{ N}$$

$$F_{\text{flat vertical}} = \frac{0.1}{2} \times 10^3 \times 10 \times 0.01 = 5 \text{ N}$$

$$F_{\text{inclined}} = 0.1 \times 0.1\sqrt{2} \times 10^3 \times 10 \times \frac{0.1}{2} = 5\sqrt{2} \text{ N}$$

$$F_{\text{triangle}} = \frac{1}{2} \times 0.1 \times 0.1 \times 10 \times \frac{2}{3} \times 10^3 \times 0.1 = \frac{10}{3} \text{ N}$$

10. Ans (A)

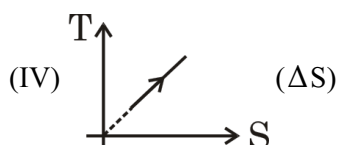
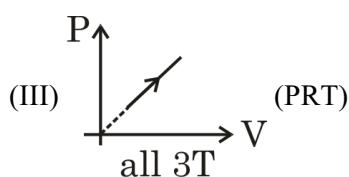
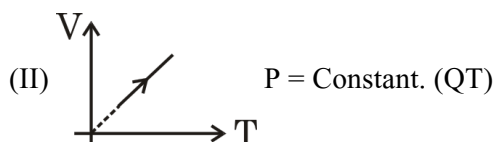
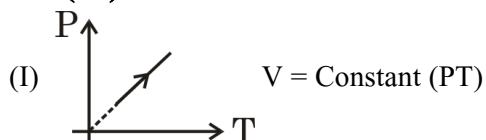
$$(I) \rho_L = \rho_b \Rightarrow \alpha \text{ \& } a = 0$$

$$(II) \rho_L > \rho_b \Rightarrow a \uparrow \alpha = 0$$

$$(III) \rho_L < \rho_b \Rightarrow a \downarrow \alpha = 0$$

$$(IV) \rho_L = \rho_b \Rightarrow a \text{ \& } \alpha = 0$$

11. Ans (B)



$$CdT = dS = \frac{dQ}{T} = nC_v \frac{dT}{T} + \frac{PdU}{T}$$

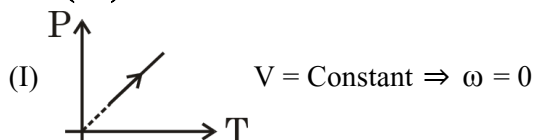
$$\left(C - \frac{C_v}{T}\right) dT = \frac{PdV}{T}$$

$dT > 0$, dU can be +ve or - ve.

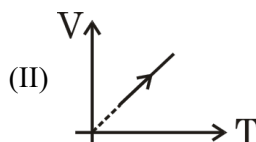
$$\left(C - \frac{C_v}{T}\right) dT = nR \frac{dV}{V}$$

$$C(T - T_0)C_v \ln\left(\frac{T}{T_0}\right) = nR \ln\frac{V}{V_0}$$

12. Ans (D)



$$\Delta U > 0 \Rightarrow \Delta H > 0 \text{ (RT)}$$

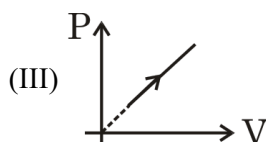


$P = \text{Constant}$

$$\Delta U > 0$$

$$\omega > 0$$

$$H > 0 \text{ (P,R,T)}$$



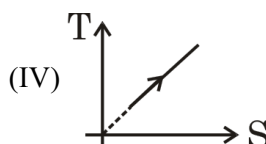
$P \propto V \Rightarrow \text{Polytropic}$

$$C = \frac{3R}{2} + \frac{R}{2} = 2R$$

$$\Delta U > 0$$

$$\omega > 0$$

$$H > 0 \text{ (P,R,T)}$$



$$\Delta S > 0 \Rightarrow H > 0$$

$$\Delta T > 0 \Rightarrow \Delta U > 0$$

ω can't be said.

PART-1 : PHYSICS

SECTION-II

1. Ans (4.00)

$$\text{COP}_1 = \frac{T_L}{T_H - T_L} = \frac{\frac{dQ}{dt} \text{ extr.}}{N_1} = \frac{C(T_H - T_L)}{N_1}$$

$$N_1 = \frac{C(T_H - T_L)^2}{T_L} = C \times \frac{100}{290}$$

$$\text{COP}_2 = \frac{T_H}{T_H - T_L} = \frac{\frac{dQ}{dt} \text{ pumped}}{N_2} = \frac{C(T_H - T_L)}{N_2}$$

$$N_2 = C \times \frac{(T_H - T_L)^2}{T_H} = C \times \frac{400}{290}$$

$$\frac{N_2}{N_1} = 4.00$$

2. Ans (330.00)

$$\pi x^2 \ell \times \rho = -k \times 2\pi x \ell \frac{dT}{dx}$$

$$\int_{r_0/2}^{r_0} x dx = \frac{-2k}{\rho} \int_{T_1}^{T_s} dT$$

$$\frac{1}{2} \left[r_0^2 - \frac{r_0^2}{4} \right] = \frac{2k}{\rho} [T_s - T_1]$$

$$\frac{1}{4} \times \frac{3}{4} \times 0.04 = \frac{-2 \times 5}{20 \times 10^3} [300 - T_1]$$

$$T_1 = 315 \text{ K}$$

3. Ans (10.00)

$$\frac{dT}{dt} = -C(T - T_0)$$

$$\ell \ln(T - T_0) \Big|_{T_1}^{T_2} = -Ct$$

$$\ell \ln \left(\frac{T_2 - T_0}{T_1 - T_0} \right) = -Ct$$

$$\ell \ln \frac{60}{30} = -C \times 5$$

$$\ell \ln \left(\frac{40}{10} \right) = -C \times t$$

$$\Rightarrow t = 10 \text{ min.}$$

4. Ans (44.50)

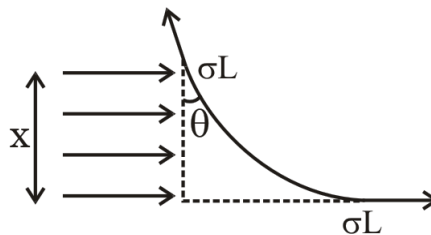
$$\frac{dH}{dt} = 1 \times \frac{4200 \times 10}{42} = 1 \text{ kW}$$

$$H = 3 \times 330 \times 10^3 + 4200 \times 4 \times 100$$

$$= 10^3 \text{ t}$$

$$t = 660 + 168 = 828 \text{ min.}$$

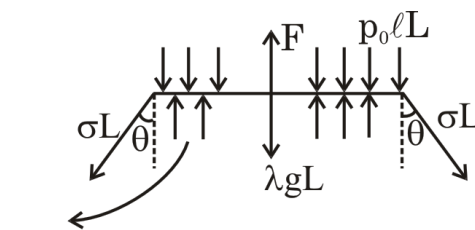
5. Ans (2.38)



$$\left(p_0 - \rho g \frac{x}{2} \right) \times xL + \sigma L = \sigma L \sin \theta + p_0 xL$$

$$\sin \theta = 1 - \frac{\rho g x^2}{2\sigma}$$

$$= 1 - \frac{10^4 \times 4 \times 10^{-6}}{2 \times 0.5} = \frac{24}{25}$$



$$(p_0 - x\rho g)\rho L$$

$$\rho_0 \ell L + \lambda g L + 2\sigma L \cos \theta$$

$$= F + (p_0 - x\rho g) \ell L$$

$$F = \lambda g L + x\rho g \ell L + 2\sigma L \cos \theta$$

$$\frac{F}{L} = \lambda g + x\rho g \ell + 2\sigma \sqrt{1 - \sin^2 \theta}$$

$$= 10 \times 10^{-3} \times 10 + 2 \times 10^{-3} \times 10^4 \times 0.1 +$$

$$2 \times 0.5 \times \frac{7}{25}$$

$$= 0.1 + 2 + 0.28 = 2.38 \text{ N/m}$$

6. Ans (1.00)

$$L_i = L_f = 0$$

$$dm (v - r\omega) = 0$$

$$\Rightarrow \omega = \frac{v}{r}$$

$$Av = \frac{d\theta}{dt}$$

$$\omega = \frac{1}{rA} \frac{d\theta}{dt} = \frac{200 \times 10^{-6}}{4 \times 10 \times 10^{-4} \times \frac{1}{20}}$$

$$\omega = \frac{2 \times 20}{40} = 1 \text{ rad/s}$$

PART-2 : CHEMISTRY

SECTION-I (i)

3. Ans (B,C)

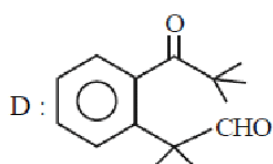
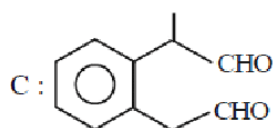
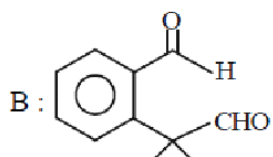
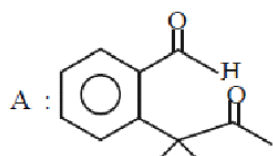
(A) Acetic acid on treatment with X_2 and NaOH undergo acid base reaction

(B) Schiff's reagent and schiff's base has different structure

(C) Aromatic aldehyde do not respond to fehling's test

(D) Carbonyl compound give 2,4-DNP test.

6. Ans (A,C)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (5.00)

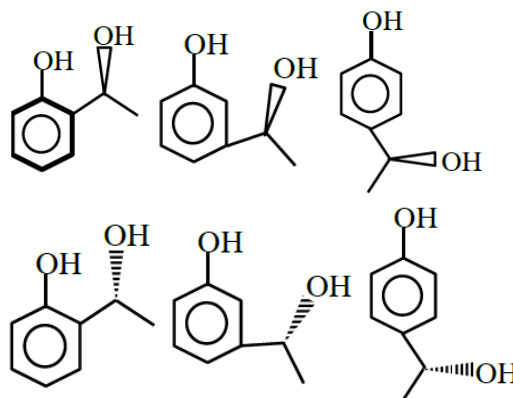
i, ii, iii, iv, v

2. Ans (6.00)

$C_8H_{10}O_2 \rightarrow$ Gives $FeCl_3$ test means Phenol derivative



Rotate plane polarized light means optically active



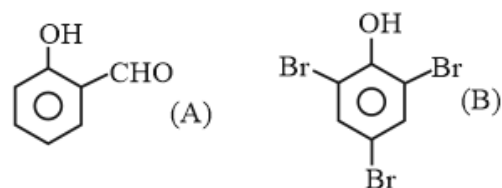
4. Ans (6.00)

a, b, c, d, e, f

5. Ans (4.00)

i, iii, iv, v

6. Ans (5.00)



PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,D)

α and β satisfy $x \cos \theta + y \sin \theta = 2a$

$$\Rightarrow (x^2 + y^2) \cos^2 \theta - 4ax \cos \theta + (4a^2 - y^2) = 0$$

$$\cos \alpha + \cos \beta = \frac{4ax}{x^2 + y^2}, \cos \alpha \cdot \cos \beta = \frac{4a^2 - y^2}{x^2 + y^2}$$

$$2 \sin\left(\frac{\alpha}{2}\right) \sin\left(\frac{\beta}{2}\right) = 1 \Rightarrow 4 \sin^2\left(\frac{\alpha}{2}\right) \sin^2\left(\frac{\beta}{2}\right) = 1$$

$$\Rightarrow \cos \alpha + \cos \beta = \cos \alpha \cdot \cos \beta$$

Hence (B)

$$\Rightarrow \frac{4ax}{x^2 + y^2} = \frac{4a^2 - y^2}{x^2 + y^2}$$

$$y^2 = 4a(a - x) \text{ Hence (A)}$$

2. Ans (A,B)

$$2 \sin \alpha \cos \alpha = 2 \cos^2 \beta; \quad \sin 2\alpha = 1 + \cos 2\beta;$$

$$\therefore \cos 2\beta = -(1 - \sin 2\alpha)$$

$$= -\left(1 - \cos\left(\frac{\pi}{2} - 2\alpha\right)\right) = -2 \sin^2\left(\frac{\pi}{4} - \alpha\right)$$

$$= -2 \cos^2\left(\frac{\pi}{4} + \alpha\right)$$

3. Ans (C,D)

$$E = \frac{1}{\sqrt{b-a}} \cdot \frac{\sqrt{b-a} \sin x}{\sqrt{\left[a + a\left(\frac{b-a}{a} \sin^2 x\right)\right]}} \cdot \sqrt{a + b \tan^2 x}$$

$$= \frac{\sin x}{|\cos x|}; \Rightarrow (C) \text{ is correct; at } x = \frac{11\pi}{4},$$

$$\frac{\sin x}{|\cos x|} = 1 \Rightarrow (D) \text{ is correct}$$

4. Ans (A,C,D)

$$\sin x \cos 20^\circ + \cos x \sin 20^\circ = 2 \sin x \cos 40^\circ$$

$$\Rightarrow \sin 20^\circ \cos x = \sin x (2 \cos 40^\circ - \cos 20^\circ)$$

$$\Rightarrow \tan x = \frac{\sin 20^\circ}{2 \cos 40^\circ - \cos 20^\circ} = \frac{\sin 20^\circ}{\cos 40^\circ + \cos 40^\circ - \cos 20^\circ}$$

$$= \frac{\sin 20^\circ}{\cos 40^\circ + 2 \sin 30^\circ \sin(-10^\circ)}$$

$$= \frac{\sin 20^\circ}{\sin 50^\circ - \sin 10^\circ} = \frac{\sin 20^\circ}{2 \cos 30^\circ \sin 20^\circ}$$

$$\Rightarrow \tan x = \frac{1}{\sqrt{3}} \Rightarrow x = 30^\circ$$

5. Ans (A,B,D)

$$\cos \alpha + \cos \beta = a$$

$$2 \cos^2 \alpha + 2 \cos^2 \beta = b + 2$$

$$\Rightarrow 1 + \cos 2\alpha + 1 + \cos 2\beta = b + 2$$

$$\cos 2\alpha + \cos 2\beta = b$$

$$\cos 3\alpha + \cos \beta = c \quad \text{---(1)}$$

$$\cos \alpha + \cos 3\beta = a$$

$$\cos^3 \alpha + \cos^3 \beta + 3 \cos \alpha \cos \beta (a) = a^3$$

$$\frac{\cos 3\alpha + 3 \cos \alpha}{4} + \frac{\cos 3\beta + 3 \cos \beta}{4}$$

$$+ 3 \cos \alpha \cos \beta (a) = a^3$$

Now use equation (1)

$$(A) (\cos \alpha + \cos \beta)^2 = \cos^2 \alpha + \cos^2 \beta + 2 \cos \alpha \cos \beta$$

$$a^2 = \frac{b+2}{2} + 2 \cos \alpha \cos \beta$$

$$\cos \alpha \cos \beta = \frac{a^2}{2} - \frac{b+2}{4}$$

$$(B) a = \cos \alpha + \cos \beta$$

$$b = \cos 2\alpha + \cos 2\beta$$

$$c = \cos 3\alpha + \cos 3\beta$$

$$2(\cos \alpha + \cos \beta)^3 + \cos 3\alpha + \cos 3\beta$$

$$= 2 \cos^3 \alpha + 2 \cos^3 \beta + 6 \cos^2 \beta + 6 \cos \alpha \cos^2 \beta$$

$$+ 4 \cos^3 \alpha - 3 \cos \alpha + 4 \cos^3 \beta - 3 \cos \beta$$

$$= 6(\cos^3 \alpha + \cos^3 \beta) + 3a(2 \cos \alpha \cos \beta - 1)$$

$$= 6[a^3 - 3a(\cos \alpha \cos \beta)] + 6a \cos \alpha \cos \beta - 3a$$

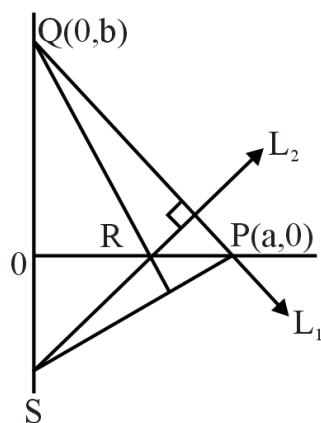
$$= 6a^3 - 12\left(\frac{a^2}{2} + \frac{b+2}{4}\right) - 3a$$

$$6a^3 - 6a^3 + 3ab + 6a - 3a$$

$$= 3a(b + 1)$$

6. **Ans (A,B,C)**

Let the equation of L_1 be $\frac{x}{a} + \frac{y}{b} = 1$, so that the coordinates of P and Q are (a, 0) and (0, b) respectively. Now, the slope of L_1 is $-\frac{b}{a}$ and therefore, that of L_2 is $\frac{a}{b}$. Let the equation of L_2 be $\frac{x}{bk} - \frac{y}{ak} = 1$



Then the coordinates of R and S are (bk, 0) and (0, -ak) respectively. Also, the equation of PA is $\frac{x}{a} - \frac{y}{ak} = 1$, and that of RQ is $\frac{x}{bk} + \frac{y}{b} = 1$. Therefore, eliminating k, we get the required locus as

$$\frac{x-a}{y} - \frac{b-y}{x} = 0$$

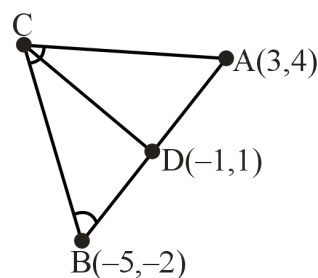
$$\Rightarrow x(x-a) + y(y-b) = 0$$

$$\Rightarrow x^2 + y^2 - ax - by = 0$$

Which is a circle passing through the origin, points P and Q.

7. **Ans (B,D)**

D(-1,1) is mid-point of AB



$$CD = BD \cot \frac{C}{2} = 5 \cot \frac{C}{2}$$

Now, $\tan C = 2$

$$\Rightarrow \frac{2 \tan \frac{C}{2}}{1 - \tan^2 \frac{C}{2}} = 2$$

$$\Rightarrow \tan^2 \frac{C}{2} + \tan \frac{C}{2} - 1 = 0$$

$$\Rightarrow \tan \frac{C}{2} = \frac{-1 + \sqrt{5}}{2}$$

$$\Rightarrow \cot \frac{C}{2} = \frac{\sqrt{5} + 1}{2}$$

$$\Rightarrow CD = 5 \left(\frac{\sqrt{5} + 1}{2} \right) = \frac{5}{2} (\sqrt{5} + 1)$$

Slope of BD is $\frac{3}{4}$

$\therefore$ Slope of CD is $-\frac{4}{3} = \tan \alpha$, so $\cos \alpha = \frac{-3}{5}$,
 $\sin \alpha = \frac{4}{5}$

Using parametric form of straight line,

coordinates of C are given by

$$\left(-1 \pm \frac{5\sqrt{5} + 1}{2} \cos \alpha, 1 \pm \frac{5(\sqrt{5} + 1)}{2} \sin \alpha \right)$$

$$\text{or } \left(-\frac{5 + 3\sqrt{5}}{2}, 2\sqrt{5} + 3 \right) \text{ and}$$

$$\left(\frac{3\sqrt{5} + 1}{2}, -(2\sqrt{5} + 1) \right)$$

8. **Ans (A,B)**

Let P, Q be (2, 0) and (6, 0) respectively

PART-3 : MATHEMATICS

SECTION-I (ii)

9. Ans (A)

$$(2x_1x_2 + 2x_2x_3 + 2x_3x_1 + x_1^2 + x_2^2 + x_3^2) + (y_1^2 + y_2^2 + y_3^2 + 2y_1y_2 + 2y_2y_3 + 2y_3y_1) = 0$$

$$\Rightarrow (x_1 + x_2 + x_3)^2 + (y_1 + y_2 + y_3)^2 = 0$$

$$\Rightarrow x_1 + x_2 + x_3 = 0$$

$$\text{and } y_1 + y_2 + y_3 = 0$$

$\Rightarrow$ Circumcentre and centroid of $\triangle ABC$ coincide

$\Rightarrow \triangle ABC$ is equilateral

$$(I) (PA)^2 + (PB)^2 + (PC)^2$$

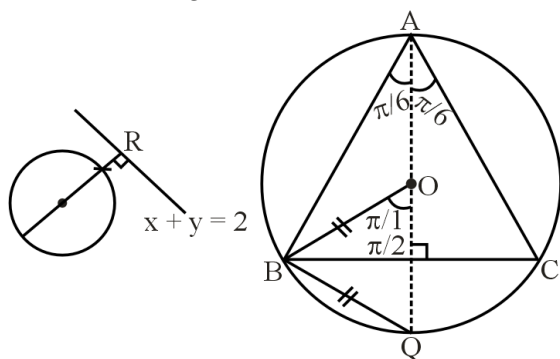
$$= (x - x_1)^2 + (y - y_1)^2$$

$$+ (x - x_2)^2 + (y - y_2)^2$$

$$+ (x - x_3)^2 + (y - y_3)^2$$

$$= 3(x^2 + y^2 + 1) = 6$$

$$(II) \angle OBQ = \frac{\pi}{3}, k = 3$$



(III) Max. distance of R from S, $d =$

$$\Rightarrow d^2 = 3 + 2\sqrt{2}$$

$$a + b = 3 + 2 = 5$$

(IV) I and G coincide with origin

$$IA = IB = IC = GA = GB = GC = 1$$

$$IA + IB + IC + GA + GB + GC = 6$$

10. Ans (D)

$$(2x_1x_2 + 2x_2x_3 + 2x_3x_1 + x_1^2 + x_2^2 + x_3^2)$$

$$+ (y_1^2 + y_2^2 + y_3^2 + 2y_1y_2 + 2y_2y_3 + 2y_3y_1) = 0$$

$$\Rightarrow (x_1 + x_2 + x_3)^2 + (y_1 + y_2 + y_3)^2 = 0$$

$$\Rightarrow x_1 + x_2 + x_3 = 0$$

$$\text{and } y_1 + y_2 + y_3 = 0$$

$\Rightarrow$ Circumcentre and centroid of $\triangle ABC$ coincide

$\Rightarrow \triangle ABC$ is equilateral

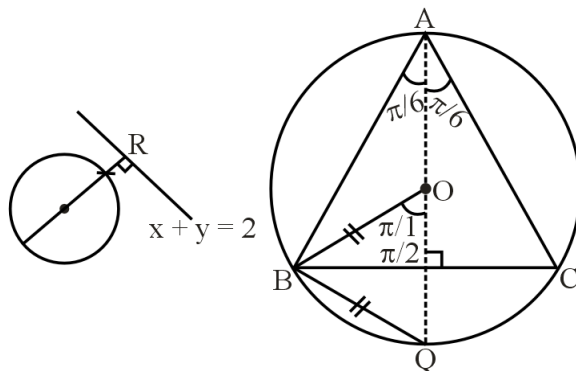
$$(I) (PA)^2 + (PB)^2 + (PC)^2$$

$$= (x - x_1)^2 + (y - y_1)^2 + (x - x_2)^2$$

$$+ (y - y_2)^2 + (x - x_3)^2 + (y - y_3)^2$$

$$= 3(x^2 + y^2 + 1) = 6$$

$$(II) \angle OBQ = \frac{\pi}{3}, k = 3$$



(III) Max. distance of R from S, $d =$

$$\Rightarrow d^2 = 3 + 2\sqrt{2}$$

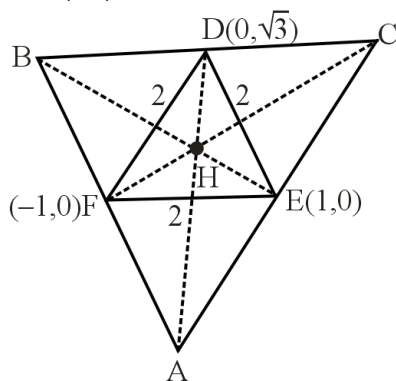
$$a + b = 3 + 2 = 5$$

(IV) I and G coincide with origin

$$IA = IB = IC = GA = GB = GC = 1$$

$$IA + IB + IC + GA + GB + GC = 6$$

11. Ans (A)



Point A, B, C are excentres w.r.t to $\triangle DEF$,

A is $(0, -\sqrt{3})$, B is $(-2, \sqrt{3})$, C is $(2, \sqrt{3})$

Orthocentre H of $\triangle ABC$ is the incentre of

$\triangle DEF$

$$\Rightarrow H = \left(0, \frac{1}{\sqrt{3}}\right)$$

(P) In radius of $\triangle ABC$,

$$r_{ABC} = \frac{2}{\sqrt{3}}$$

$$\frac{r_{ABC}}{r_{DEF}} = 2$$

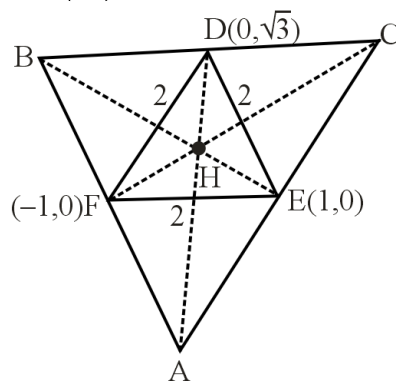
$$(Q) (AH)^2 = \left(\frac{1}{\sqrt{3}} + \sqrt{3}\right)^2 = \frac{16}{3}$$

$$\Rightarrow [(AH)^2] = 5$$

$$(R) (y_A + y_B + y_C)^2 = (\sqrt{3})^2 = 3$$

$$(S) AB = 4$$

12. Ans (C)



Point A, B, C are excentres w.r.t to $\triangle DEF$,

A is $(0, -\sqrt{3})$, B is $(-2, \sqrt{3})$, C is $(2, \sqrt{3})$

Orthocentre H of $\triangle ABC$ is the incentre of $\triangle DEF$

$$\Rightarrow H = \left(0, \frac{1}{\sqrt{3}}\right)$$

(I) In radius of $\triangle ABC$,

$$r_{ABC} = \frac{2}{\sqrt{3}}$$

$$\frac{r_{ABC}}{r_{DEF}} = 2$$

$$(II) (AH)^2 = \left(\frac{1}{\sqrt{3}} + \sqrt{3}\right)^2 = \frac{16}{3}$$

$$\Rightarrow [(AH)^2] = 5$$

$$(III) (y_A + y_B + y_C)^2 = (\sqrt{3})^2 = 3$$

$$(IV) AB = 4$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2.00)

$$\sin(2\sin x) = \sin\left(\frac{\pi}{2} - 2\cos x\right)$$

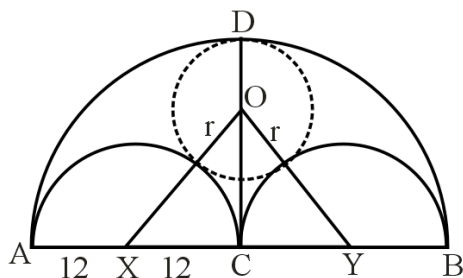
$$\sin x + \cos x = \frac{\pi}{4} ; \text{ Squaring both sides } 1 + \sin 2x$$

$$= \frac{\pi^2}{16} ; \sin 2x = \frac{\pi^2 - 16}{16}$$

$$\therefore \tan x + \cot x = \frac{2}{\sin 2x} = \frac{2 \times 16}{\pi^2 - 16} = \frac{32}{\pi^2 - 16}$$

$$\therefore a = 32, b = 16, = 2 ; \frac{a+b+c}{25} = 2$$

2. Ans (8.00)



In ΔXCO , we have

$$OX^2 = XC^2 + OC^2$$

$$\Rightarrow (12 + r)^2 = 12^2 + (24 - r)^2$$

$$\Rightarrow r = 8$$

3. Ans (2.00)

Put $x = r\cos\theta$, $y = r\sin\theta$ in given equation

$$5r^2 - 3r^2\sin 2\theta = 4$$

$$\Rightarrow r^2 = \frac{4}{5 - 3\sin 2\theta}$$

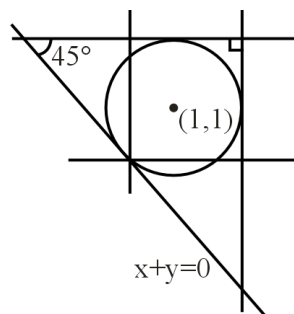
$$\Rightarrow r_{\min} = \frac{1}{\sqrt{2}} \text{ at } 2\theta = \frac{3\pi}{2}, \frac{7\pi}{2}; \text{ as } [2\theta \in (0, 4\pi)]$$

$$\text{So } \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$\text{So points are } \left(\frac{1}{\sqrt{2}}\cos\theta, \frac{1}{\sqrt{2}}\sin\theta \right)$$

$$= \left(-\frac{1}{2}, \frac{1}{2} \right) \text{ and } \left(\frac{1}{2}, -\frac{1}{2} \right)$$

4. Ans (0.00)



$$(h^2 + k^2 - 4)^2 = (h^2 + k^2 - 4h)(h^2 + k^2 - 4k) + 16$$

$$\Rightarrow (h + k)(h^2 + k^2 - 2h - 2k) = 0$$

$$\text{Circle } C_1 : x^2 + y^2 - 2x - 2y = 0 \text{ and}$$

$$C_2 : x + y = 0$$

Circumcentre will be mid point of hypotenuse in right angle triangle $(\alpha, \beta) = (0,0)$

5. Ans (5.00)

Center of the smallest circle is A

Center of the largest circle is B

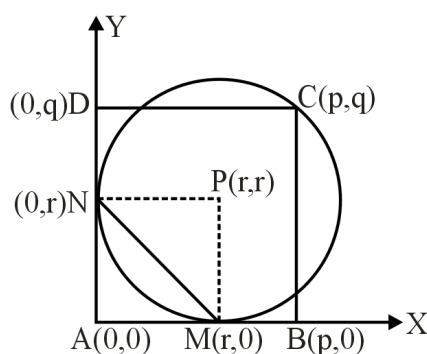
$$r_1 = |CP + CB| = 3\sqrt{2} + 3 \text{ and}$$

$$r_2 = |CP - CB| = 3\sqrt{2} - 3$$

$$\frac{r_1}{r_2} = \frac{3\sqrt{2} + 3}{3\sqrt{2} - 3} = 3 + 2\sqrt{2}$$

6. Ans (25.00)

Let us take AB and AD as coordinate axes



If r be the radius of circle, then its centre is $P(r, r)$.

$$\text{Equation of circle is } (x - r)^2 + (y - r)^2 = r^2$$

$$\Rightarrow x^2 + y^2 - 2rx - 2ry + r^2 = 0$$

Let the coordinates of $C = (p, q)$

Equation of MN is $x + y = r$

$$\text{Its distance from C is 5 units } \Rightarrow \frac{|p + q - r|}{\sqrt{2}} = 5$$

$$\Rightarrow (p + q - r)^2 = 50$$

Since (p, q) lie on the circle,

$$p^2 + q^2 - 2rp - 2rq + r^2 = 0$$

$$\Rightarrow (p + q - r)^2 - 2pq = 0$$

$$\Rightarrow 50 - 2pq = 0 \Rightarrow pq = 25$$

Area of rectangle = 25 sq. units